

Pharmacodynamics

Pharmacodynamics is the study of the biochemical and physiological effects of drugs and their mechanisms of action.

- Deals with what does the drug on the body

Drug *receptor* or drug *target* is the cellular macromolecule or macromolecular complex with which the drug interacts to elicit a cellular response.

Types of receptor

Intracellular receptors: These are protein receptors that require the drug to cross the plasma membrane;

- The drug needs to be lipophilic. E.g Steroids

Transmembrane enzymes: A drug binds to the extracellular component of the receptor then activates an enzymatic reaction in the intracellular component.

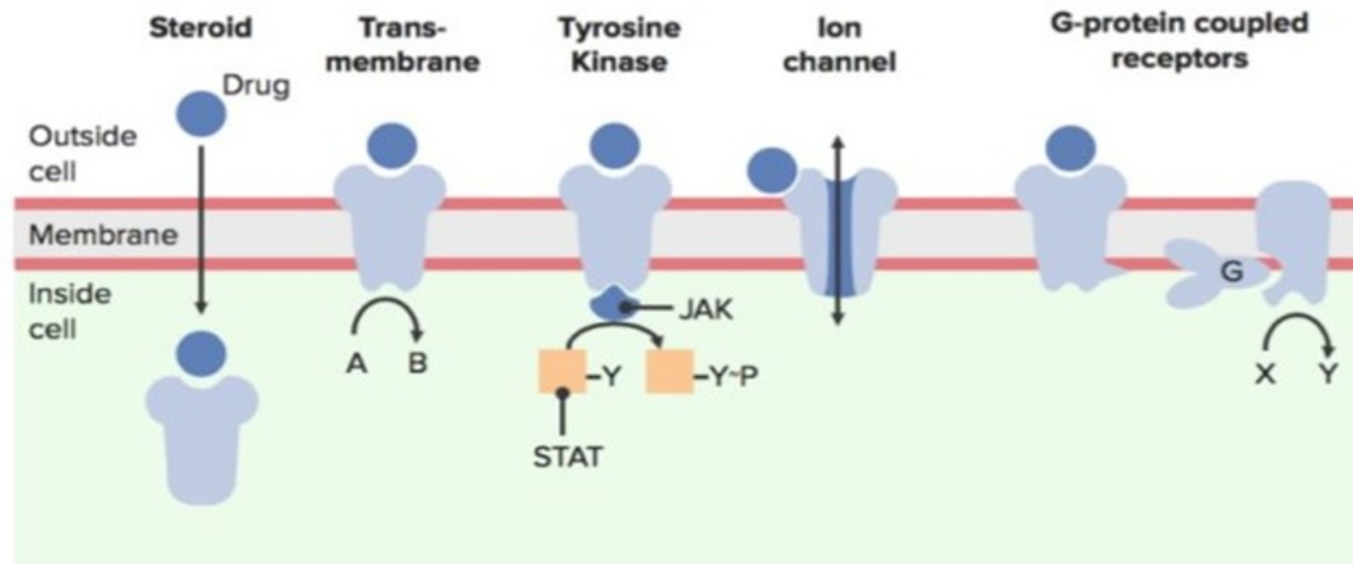
Tyrosine kinase: When a drug binds to the extracellular component of this receptor, it leads to dimerizing of the two parts of the receptor intracellularly.

E.G Growth hormones

Ligand-gated ion channels: These ion channels are ligand-gated, i.e., they are closed until the receptor binds to the drug

For example, drugs that stimulate GABA receptors on the neurons cause chloride influx

Figure: Types of receptors



Agonists - Drugs that bind to physiological receptors and mimic the regulatory effects of the endogenous signaling compounds

partial agonists- Agents that are only partly as effective as agonists

Inverse agonists have a higher affinity for the inactive **R_i** state than for **R_a** and decrease or abolish any constitutive activity.

Agonist can be *primary agonist* or *Allosteric (allotropic)*

Pharmacodynamics...

primary agonist the drug binds to the same *recognition site* as the endogenous agonist

allosteric agonists bind to a different region on the receptor.

Antagonist- Drugs that block or reduce the action of an agonist

Antagonism results from competition with an agonist for the same or overlapping site on the receptor

Types of antagonism

- a) Physical antagonism
- b) Chemical antagonism
- c) Physiological/functional antagonism
- d) Pharmacological antagonism

Physical antagonism

- *Based on physical property of drugs*
e.g. charcoal (adsorb alkaloid) in alkaloidal poisoning

Chemical Antagonism

A type of antagonism where a drug counters the effect of another by simple **chemical reaction** / neutralization(not binding to the receptor)

e.g. Calcium sodium edetate form insoluble complexes with arsenic / lead

Antacids (Aluminum/magnesium hydroxide + HCL)

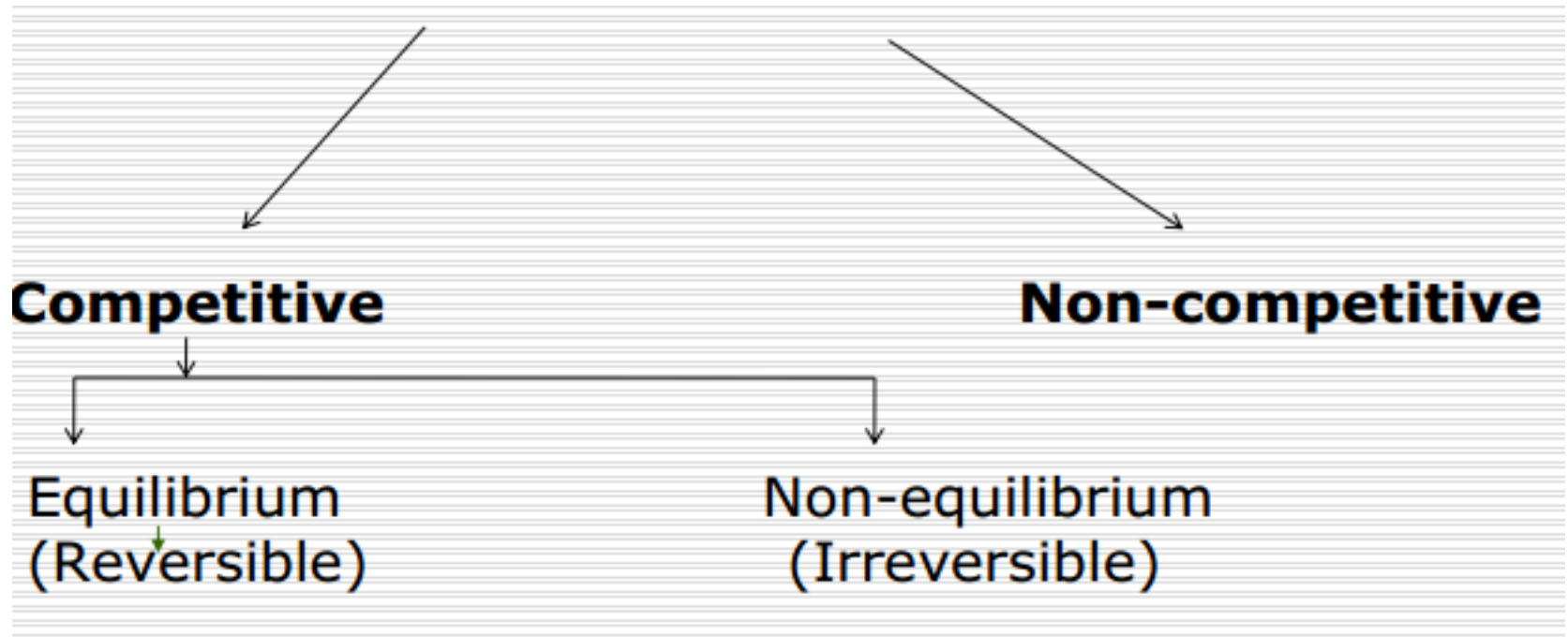
Functional Antagonism (Physiological Antagonism)

- It is Opposite effects of two drugs on same function
- Two drugs act on two diff. types of receptors & antagonize action of each other
- e.g. Glucagon and insulin on blood sugar level
β2 agonists and cholinergic drugs on respiratory smooth muscle cells

Pharmacological Antagonism

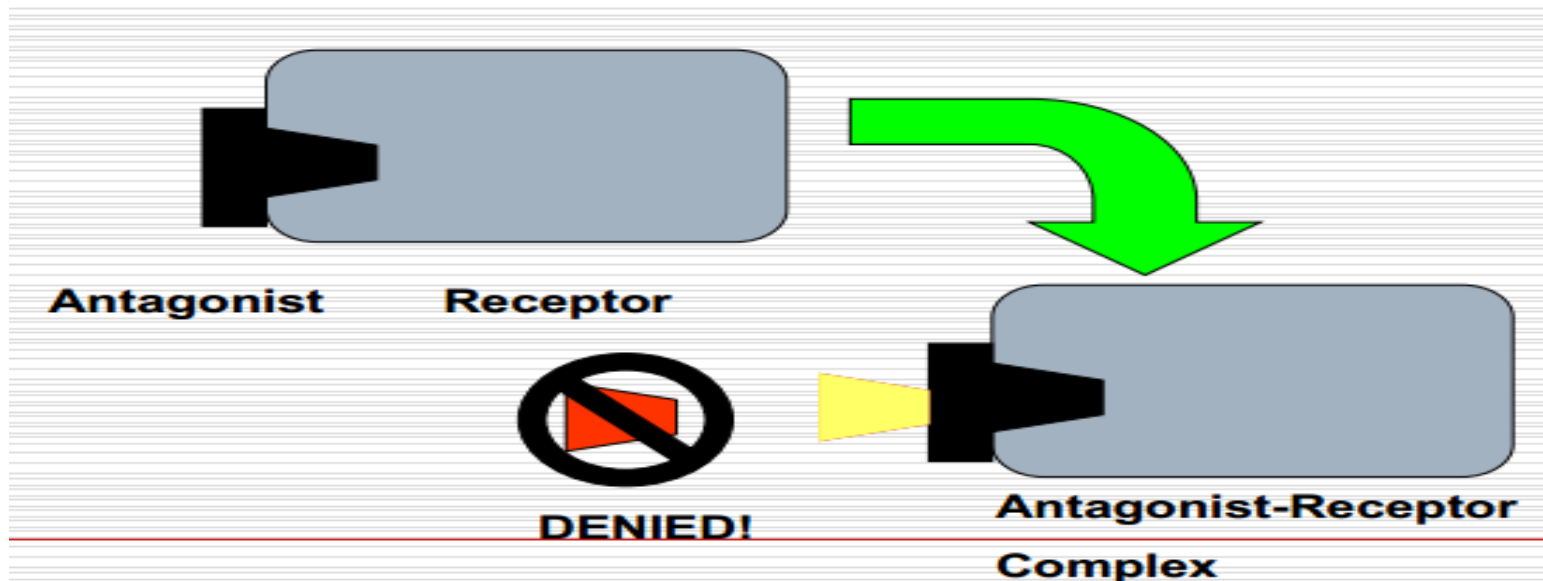
- Antagonist will prevent the effect of agonist by binding on the same receptor either on the same or d/f site of agonist binding site

Pharmacological antagonism



Competitive Antagonism

- The antagonist will bind on the same site of the receptor
- **antagonist competes** with the substrate to bind to the receptor

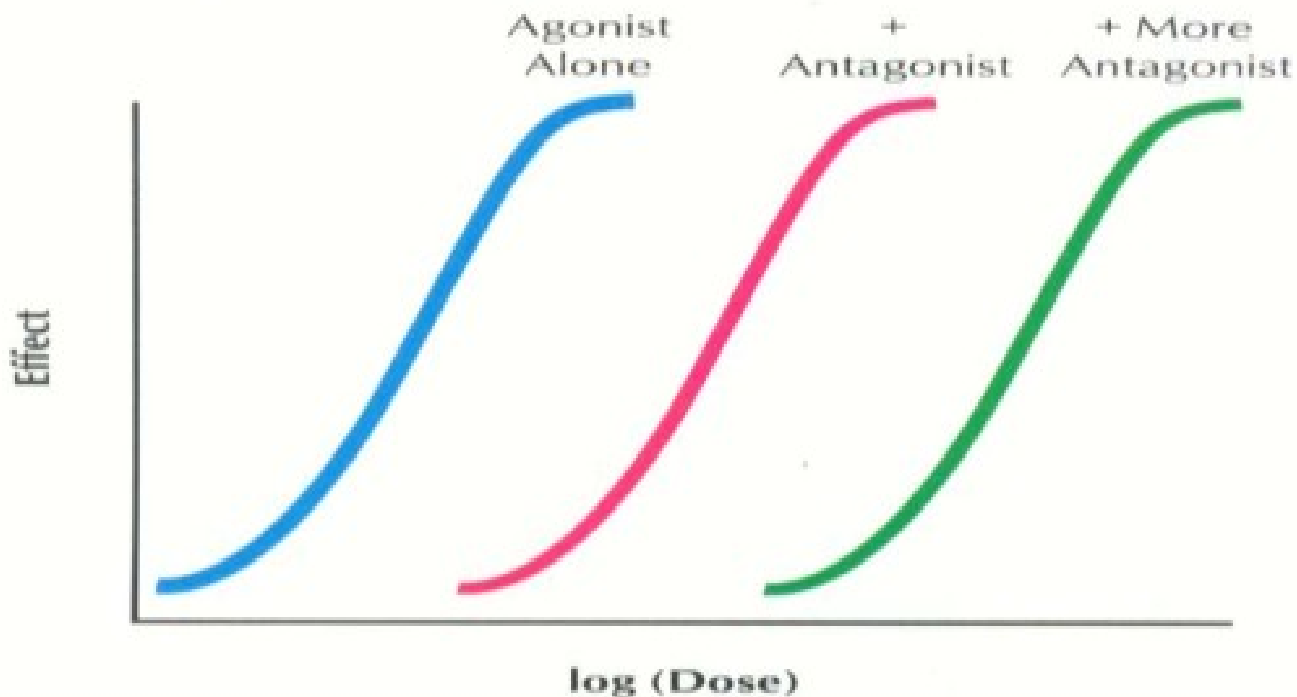


Competitive reversible antagonism

- **Antagonist will bind with the same receptor by forming Weak bonds**
- **Maximal response is concentration dependent**
- **The effect of antagonist can be overcome by increasing the concentration of agonist**

Competitive reversible antagonism

Antagonists: Surmountable (Reversible)



e.g. Atropine and Acetylcholine at Muscarinic -R
Naloxone and Morphine at opioid-R

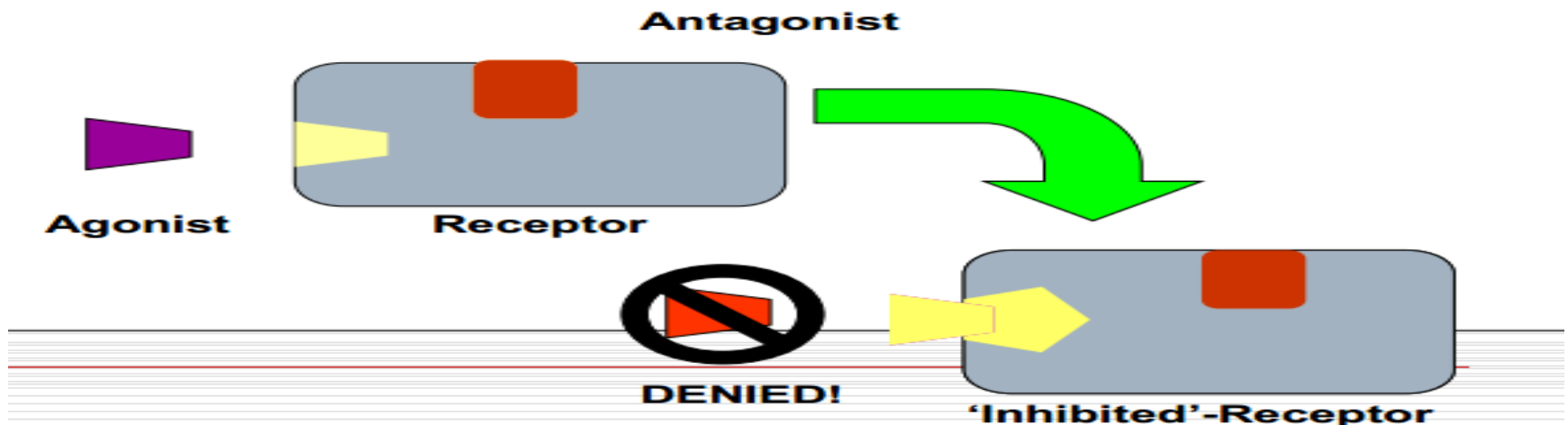
Competitive irreversible antagonism

- Have affinity for the same receptor sites and bind in an irreversible manner by **covalent bond**
- Effects **cannot** be **overcome** even by increasing the concentration of the agonist

Non competitive antagonism

- Binds to site other than the agonist site
- Prevent the receptor activation by the agonist

e.g. Flumazenil by binding to BZD site antagonizes the effects of BZD by preventing the binding of GABA to GABAA receptor



Receptor desensitization and up regulations

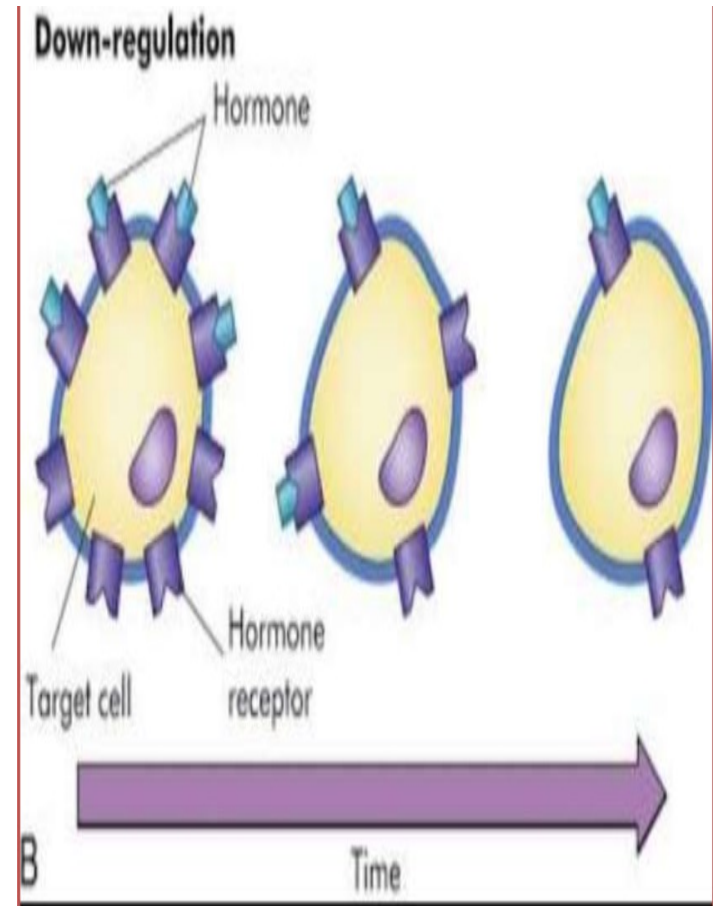
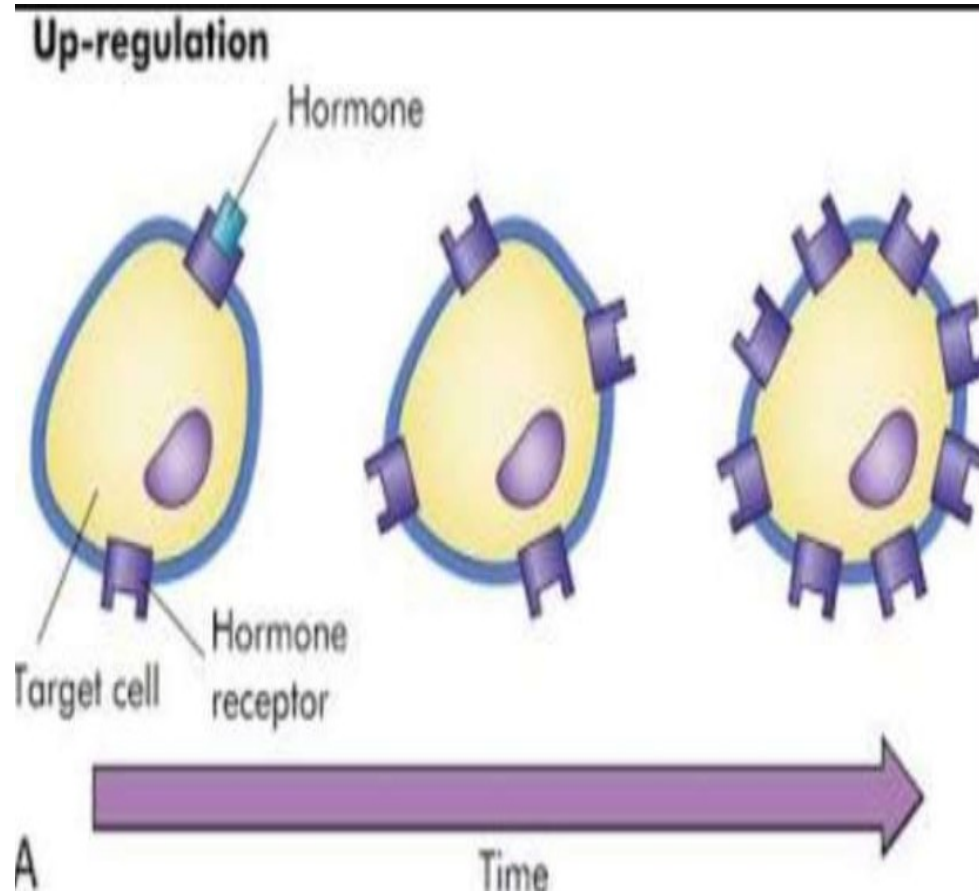
Upregulation (i.e., increase in the number) of receptors

❖ occurs when the **activity of the receptor is lower than usual** (e.g., due to long-term administration of an antagonist).

▪ E.g administration of beta-blockers upregulates β adrenoreceptors.

Downregulation: Downregulation (i.e., decrease in number) is the occurs due to repeated or long-term administration of an agonist.

Receptor down regulation and desensitization

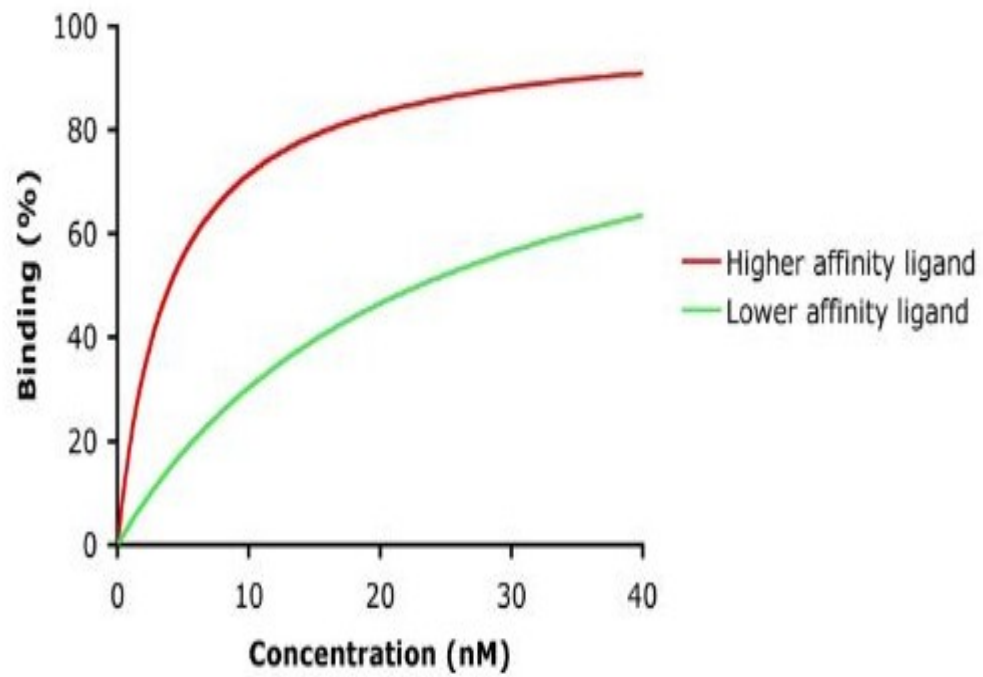


Affinity, Efficacy, and Potency

❖ **In general**, the drug-receptor interaction is characterized by (1) binding of drug to receptor and (2) generation of a response in a biological system

❖ **Affinity**

- ✓ Affinity is the measure of the strength of the bond between the drug and its receptor.
- ✓ The affinity of a drug to its receptor helps determine the dose of the drug:
 - low affinity would indicate the need of a higher dose to form enough drug-receptor complexes that would lead to a significant effect

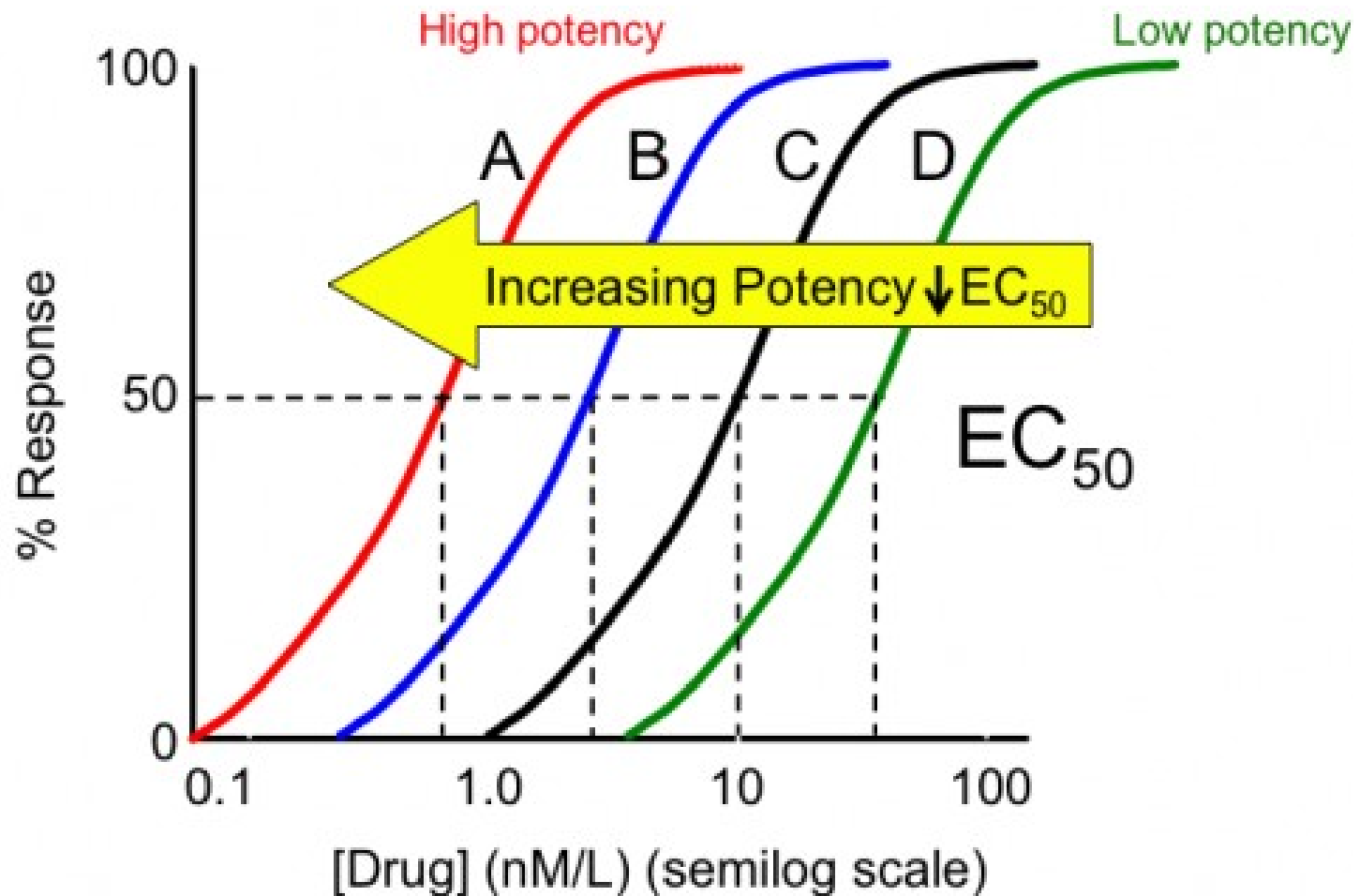


Affinity, Efficacy, and Potency...

Potency

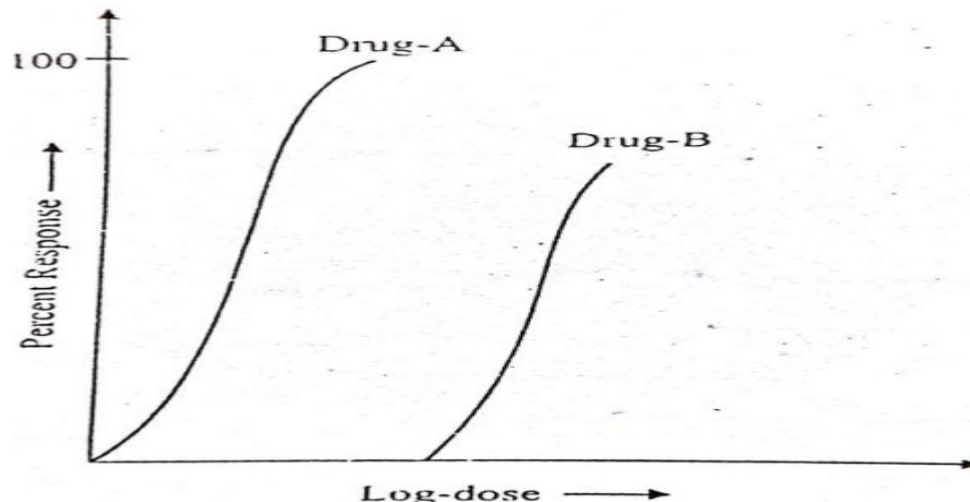
- The **quantity** of drug necessary to produce a given effect, or
- when 2 drugs produce equivalent responses, the drug which produces an effect with lowed dose is more potent.
- The concentration producing a half-maximal effect [EC_{50}] is smaller is said to be the more potent

Potency....



Efficacy

- ❖ Efficacy reflects the **capacity of a drug** to activate a receptor and generate a cellular response.
- ❖ Thus, a drug with **high efficacy** may be a full agonist, eliciting, at some concentration, a full response.
- ❖ A drug with a lower efficacy at the same receptor may not elicit a **full response** at any dose ($A > B$)

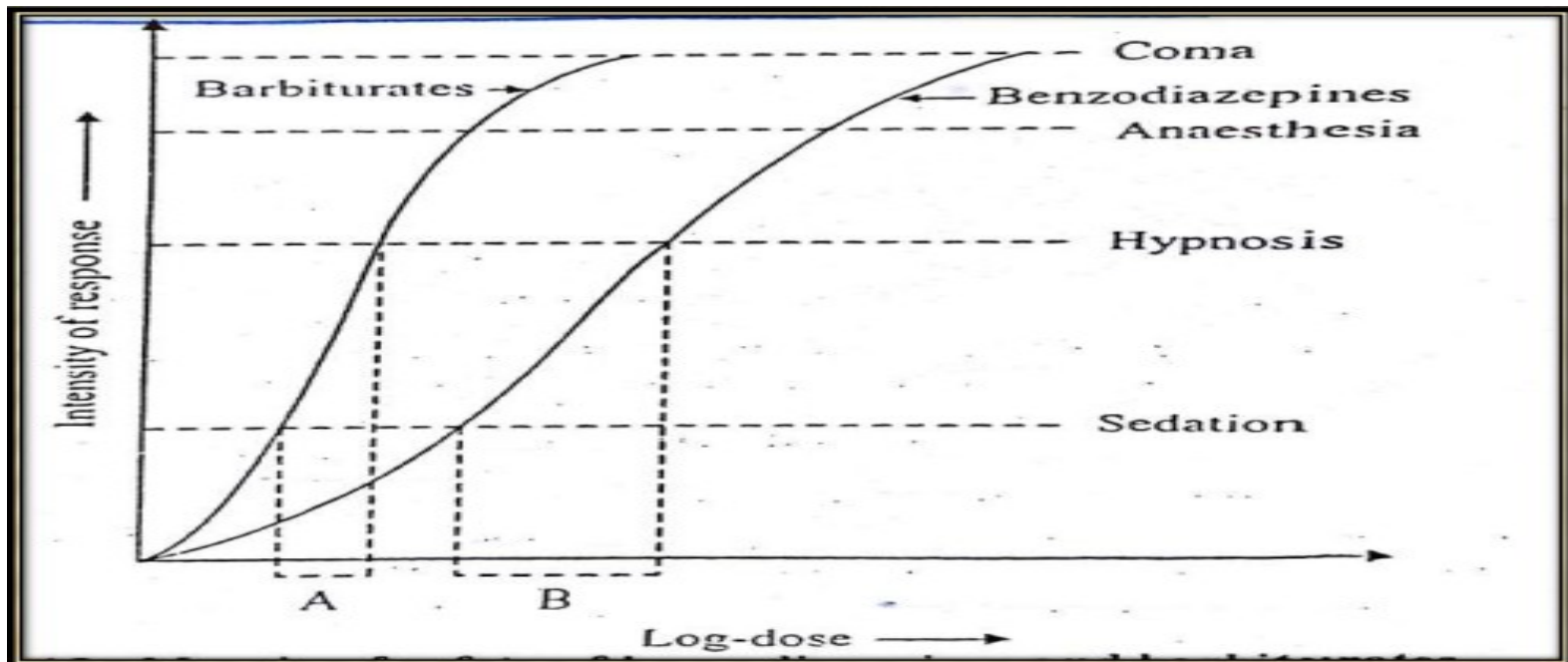


Safety

- The gap between the **therapeutic effect** DRC & the **adverse effect** DRC defines therapeutic index of a drug.
- The therapeutic index gives a measure of safety margin of a drug
- Therapeutic index is a range between the dose which produces **minimal therapeutic effect** & the dose which produces **maximal acceptable adverse effect**.
- Therapeutic index =
$$\frac{\text{Median lethal dose}}{\text{Median effective dose}}$$

Safety..

Median effective dose (ED50) is the dose of a drug necessary to produce an effect for half of all population
median lethal dose (LD50) is dose of a drug lethal to half of population



Dose-response relationships

There are two types of dose–response relationship

Graded dose-response relationships:

- ✓ a Graded dose-response relationship describes a drug effect which **increases in proportion to increasing drug dose**.
- ✓ A graded response to a drug is seen in an **individual**, and increases with dose.
- ✓ Graded dose-response graphs plot the **response to a drug against its concentration**

Graded dose-response...

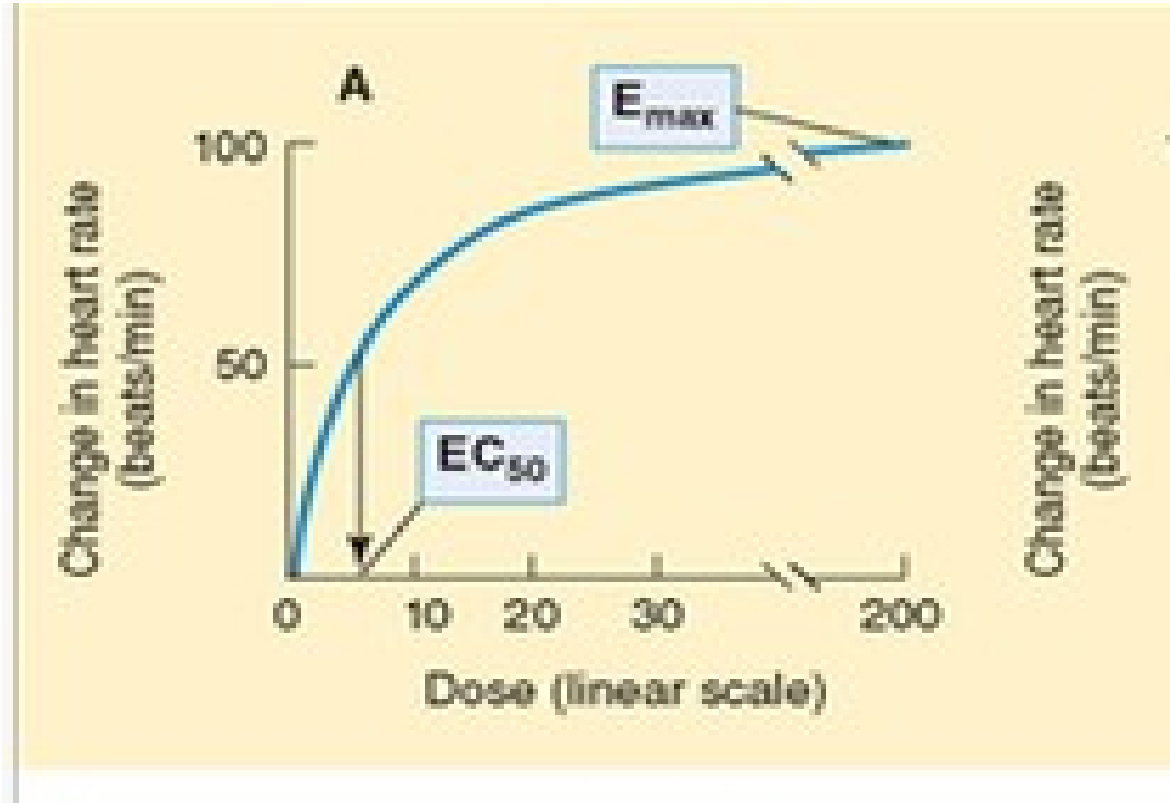


Figure: Graded dose response

Dose-response relationships

❖ **Quantal dose-response relationships:**

- ✓ describes a drug effect which is binary (either **present** or **absent**).
- ✓ A quantal response to a drug is observed in a population, and is either **present** or **absent** in any single individual.
- ✓ Quantal dose-response graphs plot the rate of an **outcome occurrence** in a population against the **drug dose**.

Quantal dose-response....

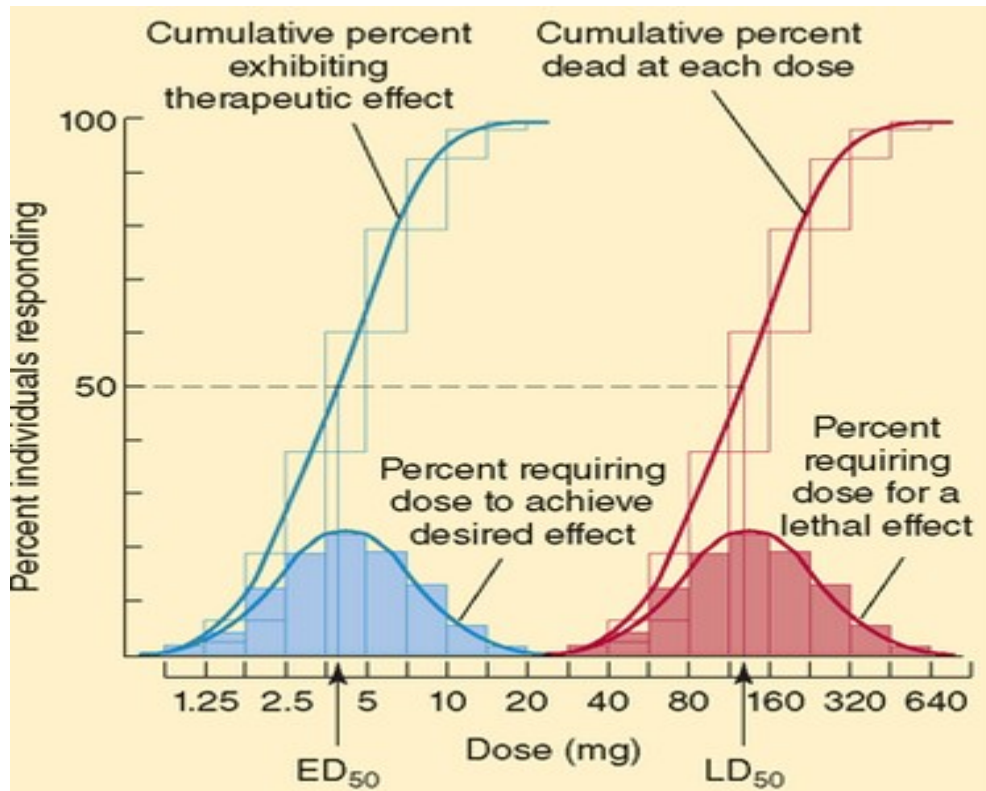


Figure: Quantal dose response

- ❖ **Adverse drug event (ADE)** is “any untoward medical occurrence that may present during treatment with a pharmaceutical product but which does not necessarily have a causal **relationship** with this treatment.
- ❖ **Adverse drug reaction (ADR)** is “a response to a drug which is noxious and unintended and which occurs at doses normally used in man for prophylaxis, diagnosis, or therapy of disease or for the modification of physiologic function.”
- ❖ An ADR is a type of ADE whose cause can be directly attributed to a drug and its physiologic properties

- ADRs occur with appropriate OR Inappropriate prescribing and dosing
- Side effect: any effect caused by drug other than the intended therapeutic effect whether beneficial, neutral or harmful

Hypersensitivity

- ❖ **Hypersensitivity** - refers to undesirable (damaging, discomfort producing and sometimes fatal) reactions produced by the normal immune system.
- ❖ Hypersensitivity reactions require a pre-sensitized (immune) state of the host.
- ❖ Based On the mechanisms involved and time taken for the reaction Hypersensitivity reactions can be divided into four types:
type I, type II, type III and type IV

Type I Hypersensitivity

- ❖ It is also known as **immediate** or **anaphylactic** hypersensitivity.
- ❖ The reaction may involve skin (**urticaria** and **eczema**), eyes (conjunctivitis), nasopharynx (rhinorrhea, rhinitis), bronchopulmonary tissues (asthma)
- ❖ The reaction takes **15-30** minutes from the time of exposure to the antigen.
- ❖ Sometimes the reaction may have a delayed onset (**10-12** hours)
- ❖ It is mediated by **IgE**

Type II Hypersensitivity

- ❖ It is also known as **cytotoxic hypersensitivity** and may affect a variety of organs and tissues.
- ❖ The antigens are normally endogenous which can attach to cell membranes can also lead to type II hypersensitivity.
- ❖ The reaction time is **minutes to hours**
- ❖ is primarily mediated by antibodies of **IgM** or **IgG**

Example:

Drug-induced **hemolytic anemia**, granulocytopenia and thrombocytopenia

Type III Hypersensitivity

- ❖ It is also known as **immune complex hypersensitivity**
- ❖ The reaction may be general
- ❖ The reaction may take **3-10 hours** after exposure to the antigen
- ❖ It is mediated by soluble immune complex (IgG and IgM)

Type IV Hypersensitivity

- ❖ It is also known as **cell mediated** or **delayed type hypersensitivity**.
- ❖ Type IV hypersensitivity is involved in the pathogenesis of many autoimmune and **infectious diseases** (tuberculosis, leprosy, histoplasmosis, toxoplasmosis, leishmaniasis, etc.)

Drug interactions

Drug interaction refers to modification of response to one drug by another when they are administered simultaneously or in quick succession”

- Drug-Drug Interactions
- Drug – Herbs Interactions (Indigenous Medicines)
- Drug-Food Interactions (Grape Fruit Juice etc.)

MECHANISM OF DRUG INTERACTIONS

- Drug interactions can be broadly divided into
 - **Pharmaceutical Interaction**
 - During dosage form preparation or at time of administrations.
 - Dissolving the drug in solvent,
 - Mixing drugs in powder, solution or injection forms.
 - **Pharmacokinetic (ADME)**
 - Absorption (Complex or Chelate formation, Altered stomach pH, Ionization, GIT motility, First Pass Metabolism)
 - Distribution (Protein binding)
 - Metabolism (Enzyme induction/inhibition)
 - Excretion (Altered pH, Ionization, Entero-hepatic recirculation)
 - **Pharmacodynamic (At receptor or tissue level)**

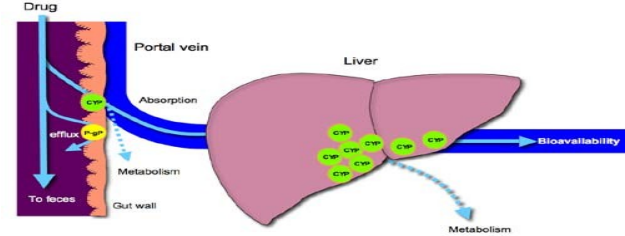
Pharmaceutical interactions:

- Also called as **incompatibility**.
- it is a **physicochemical interaction** that occurs when drugs are mixed in i.v . Infusions causing **precipitation** or **inactivation** of active principles .

Example:-

Ampicillin ,chlorpromazine & barbituates *interact with dextran in solutions and are broken down or from chemical compounds.*

ABSORPTION



- **Insoluble and poorly absorbed complexes in the gut**
 - Example:-
 - Tetracyclines and calcium/iron salts, antacids or sucralfate
 - Phenytoin absorption is decreased by sucralfate
- **Alteration in Entero-hepatic recirculation**
 - Antibiotics like Tetracyclines (Broad Spectrum) markedly **reduce gut flora** that normally deconjugates oral contraceptive steroids secreted in the bile as glucuronides and permits their **Entero-hepatic recirculation**.
Contraceptive failure when concurrent use of antibiotics due to lowering of the contraceptive blood levels.

DISTRIBUTION

- **Primarily due to displacement** of one drug from its binding sites on plasma proteins by another drug.
- Drugs highly bound to plasma proteins that have a relatively small volume of distribution
 - like oral anticoagulants, sulfonylureas, certain NSAIDs and anti-epileptics are particularly liable to displacement interactions
- The drug which is in unbound form is active while portion which is in bound form works as temporary storage.

METABOLISM

- Certain drugs **reduce or enhance the rate of metabolism** of other drugs and **affect the bioavailability**.
- Inhibition of drug metabolism may be due to competition for the same CYP450 iso-enzyme

Metabolism Interactions

1.Enzyne Induction

Corticosteroids, Oral

Contraceptives, Coumarins,

Phenytoin

Barbiturates

Decreased Plasma Levels;

Decreased Efficacy Of Object
Drugs

Oral Contraceptives, Oral

Hypoglycaemics

Rifampin

Decreased Plasma Levels

2.Enzyme Inhibition

Tyramine Rich Food

MAO Inhibitors

Enhanced Absorption Of Un
Metabolised Tyramine.

Coumarins

Metranidazole Phenyl

Butazone

Increased Anti Coagulant
Activity.

Excretion Interactions

Are these where **the excretion pattern of the object drug is altered.**

Major mechanisms of excretion interactions are-

- Alteration of urine PH
- Competition for active secretions

Excretion Interactions

1.Changes in Active Tubular Secretion

Pencillin, Cephalosp
Orins, nalidixic Acid

Probenicid

Elevated Plasma Levels
of Acidic Drugs

2.Changes in Urine PH

Amphetamine

Antacids, thiazides,
acetazolamide

Increased Passive
Reabsorption Of Basic
Drugs. increased risk of
toxicity

3.Changes in Renal Blood Flow

Lithium Bicarbonate

NSAIDS

Decreased Renal
Clearance of Lithium.
Risk Of Toxicity

PHARMACODYNAMIC INTERACTIONS

- These interactions derive from modification of the action of one drug at the target site by another drug, **independent of a change in its concentration.**
- This may result in an **enhanced response (synergism)**, an **attenuated response (antagonism)** or an abnormal response.

Direct Pharmacodynamic *Interactions*:

In which drugs having similar or opposing pharmacological effects are used concurrently.

The three consequences of direct interactions are

1. Antagonism.
2. Addition or summation
3. Synergism or potentiation.

Antagonism: The interacting drugs have opposing actions

Example: Acetylcholine and noradrenaline have opposing effects on heart rate.

Addition or summation:

The interacting drugs have similar actions and the resultant effect is the sum of individual drug responses

Example: CNS depressants like sedatives and hypnotics,...etc

Synergism or potentiation:

It is an enhancement of action of one drug by another **Example:**

Alcohol enhances the analgesics activity of aspirin.